

NANTALE LETICIA HOPE 2023-B401-10399
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PROJECT REPORT: ONLINE VOTING SYSTEM

1.0 INTRODUCTION

The Online Voting System is a web-based application developed to enable students to vote for their preferred candidates electronically. The system ensures secure, transparent, and efficient voting while preventing multiple voting by the same user. It was developed using PHP, MySQL, HTML, CSS, and JavaScript, and managed using Git and GitHub.

2.0 OBJECTIVES OF THE SYSTEM

- To enable secure online voting for students
- to eliminate manual voting errors and malpractice
- to ensure one-person-one-vote principle
- to provide real-time vote counting
- to simplify election administration

3.0 SYSTEM DESCRIPTION

The system consists of three modules: Student Module, Admin Module, and Database Module. Students vote for candidates, admin manages the system, and the database stores all records securely.

4.0 DATABASE STRUCTURE

Users Table:

- id (PK, Auto Increment)
- full name (VARCHAR)
- email (VARCHAR, UNIQUE)
- password (VARCHAR)
- role (ENUM: student, admin)

Candidates Table:

- id (PK, Auto Increment)
- name (VARCHAR)
- position (VARCHAR)
- manifesto (TEXT)
- image (VARCHAR)

Votes Table:

- id (PK, Auto Increment)
- user_id (FK)
- candidate_id (FK)
- voted at (TIMESTAMP)

5.0 SYSTEM FEATURES

- Secure login authentication
- One vote per user enforcement
- Candidate management system
- Real-time vote counting
- Responsive user interface

6.0 CHALLENGES FACED

Challenges included Git/GitHub setup issues, database connection errors, merge conflicts, and implementing one-vote restriction. These were resolved through debugging and research.

7.0 LESSONS LEARNED

Improved understanding of full-stack development, database design, Git version control, debugging, and system testing.

8.0 CONCLUSION

The system successfully provides a secure and efficient voting platform that ensures transparency and prevents fraud.